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Digital Documentation, Mapping, and the Bell Island Virtual Memorial

The Project

The Bell Island Expedition is a Canadian-led field project to document the wrecks sunk during the German submarine attacks of 1942 and to place them within their full historical, archaeological, and environmental context.

The expedition integrates underwater survey, three-dimensional photogrammetry, sonar mapping, aerial and terrestrial imaging, archival research, and marine science to record the wrecks, coastal defence sites, and the surrounding landscape as a single connected wartime battlefield.

The four ships sunk at Bell Island remain among the best-preserved Second World War wreck sites in Canadian waters. Because they lie close to shore and near surviving land-based military and industrial features, they provide a rare opportunity to study the Battle of the Atlantic as a complete landscape, including the naval battlefield, the iron ore mines that formed the target, the coastal artillery batteries, the damaged pier facilities, and the memorial sites that commemorate those who were lost.

Unlike most naval battlefields, Bell Island preserves the full physical context of war—from seabed wrecks to shoreline defences and inland mining infrastructure—within a single, coherent and accessible landscape.

The Bell Island Digital Terrain Model (DTM)

The primary objective of the expedition is to create a high-resolution Digital Terrain Model (DTM) of the Bell Island battlefield.

This model will integrate:

- underwater photogrammetry of the wrecks
- sonar mapping of the seabed
- drone-based aerial imaging
- terrestrial laser scanning of land-based features

The completed DTM will allow users to visualize, in a single environment:

- the wrecks beneath the sea
- the shoreline and harbour approaches
- coastal artillery positions
- inland industrial mining infrastructure
- and commemorative sites

This unified model transforms Bell Island from a series of individual locations into a coherent historical landscape.

Photogrammetry of the Four Wrecks

High-resolution photogrammetry will be used to document:

- *Saganaga* (British)
- *Lord Strathcona* (Canadian)
- *Rose Castle* (Canadian)
- *PLM-27* (Free French under British wartime control)

These models will allow detailed three-dimensional examination of:

- torpedo damage and blast effects
- structural collapse and site formation processes
- cargo remains and industrial context
- marine growth and ecological colonization

They will also establish a biological and ecological baseline, supporting future research into environmental change and the long-term impacts of climate change in Conception Bay.

Because the wrecks lie at different depths and are subject to varying currents and visibility conditions, and are widely separated, documentation will require multiple dive operations, remotely operated vehicles (ROVs), and advanced underwater imaging systems.

The result will be a permanent digital record of each site, preserved without physical disturbance.

Integration of Land and Sea

Unlike most Battle of the Atlantic sites, Bell Island's battlefield extends directly from seabed to shoreline.

The expedition will include documentation of:

- the Cemetery Hill coastal artillery battery
- the remains of Scotia Pier
- mine entrances and industrial structures
- the Lance Cove Seamen's Cemetery
- other surviving wartime locations

Drone photography, terrestrial laser scanning, and archival mapping will be combined with underwater survey data to produce a fully integrated land–sea model of the battlefield.

Experience from the HMCS *Canada* Expedition

The Bell Island project builds directly on the methodology developed during the HMCS *Canada* Expedition, which produced a complete three-dimensional photogrammetry model of Canada's first purpose-built warship.

That expedition demonstrated how modern imaging technology can be used to document historic shipwrecks in a way that allows them to be studied and interpreted without disturbing the site.

Examples of this work can be seen on the HMCS *Canada* Expedition Projects page:
<https://www.hmcscanadaexpedition.org/projects>

The Bell Island Expedition applies these same techniques on a larger scale, extending from a single wreck site to an entire battlefield landscape.

Experience from the 2016 NOAA Battle of the Atlantic Expedition

Members of the expedition team, including Joseph Frey and imaging specialists from VOYIS (then operating as 2G Robotics), also participated in the 2016 NOAA Battle of the Atlantic Expedition.

That expedition demonstrated the value of combining historical research, sonar mapping, and high-resolution imaging to document underwater battlefields.

The experience gained during that work helped shape the approach being applied at Bell Island.

The Bell Island Battle of the Atlantic Virtual Memorial

The Bell Island 2027 project is intended as the foundation for a larger digital documentation program that will develop into a comprehensive Battle of the Atlantic Virtual Memorial.

Information collected during the 2027 expedition—including photogrammetry, sonar data, aerial imagery, and historical research—will be combined to create a detailed interactive representation of the Bell Island battlefield.

This platform will allow users to explore the site visually and to understand how the events of 1942 unfolded within the wider system of convoy routes, industrial infrastructure, and coastal defences.

Development of the Memorial Website

Over the coming years, the Bell Island website will expand as additional material is added.

Future development may include:

- interactive maps of North Atlantic convoy routes
- three-dimensional models of additional wrecks
- historical photographs and archival documents
- cartographic reconstructions of wartime bases and infrastructure
- educational material for students and researchers

As the project grows, the site will evolve from an expedition platform into a permanent Battle of the Atlantic Memorial and research resource, linking Bell Island to other sites across Newfoundland, Canada, and the wider Atlantic world for heritage-based tourism, scholarly research, and international commemoration.

Future phases of the website will expand this foundation to include a comprehensive interpretation of the North Atlantic system and the global naval war. These sections will examine convoy routes, naval bases, air operations, and submarine campaigns across the Atlantic and beyond, placing Bell Island within the full operational and strategic framework of the Second World War.

A Long-Term Documentation Project

The Bell Island Expedition 2027 represents the first stage of a multi-year project to document and preserve one of the most important integrated naval battlefields of the Second World War, forming the foundation for a significantly expanded second phase of this website.

By combining history, maritime archaeology, marine science, and advanced imaging technology, the project will create a lasting digital record that remains accessible to researchers, students, and the public.

As this work continues, Bell Island will serve as the foundation for a broader interpretive framework connecting Canada and the North Atlantic to the global history of the Battle of the Atlantic.